# C20 Electrode Potentials and Cells – Knowledge Organiser

*AQA A-level Chemistry • Set 1: Half-Cells and the SHE* **Paper 1**

| # | Question | Answer |
|---|---|---|
| 1 | Standard electrode potential | EMF of a half-cell against the SHE under standard conditions |
| 2 | Conditions in the standard hydrogen electrode | $H_2$ at 100 kPa, 1.00 mol $dm^{-3}$ $H^+$, 298 K |
| 3 | Electrode in the SHE | Platinum |
| 4 | $E^{\ominus}$ of the SHE | 0.00 V |
| 5 | Ion concentration in a standard half-cell | 1.00 mol $dm^{-3}$ |
| 6 | Electrode used when no metal is in the half-cell | Platinum |
| 7 | Why platinum is used | Inert and conducts |
| 8 | Purpose of the salt bridge | Completes the circuit; lets ions move |
| 9 | Contents of the salt bridge | Saturated $KNO_3$ |
| 10 | Why KCl is not used with $Ag^+$ half-cells | Forms AgCl precipitate |

# C20 Electrode Potentials and Cells – Knowledge Organiser

*AQA A-level Chemistry • Set 2: Cells and Predictions* **Paper 1**

| # | Question | Answer |
|---|---|---|
| 1 | EMF equation | $E_{cell} = E_{right} - E_{left}$ |
| 2 | Symbol for a phase boundary in cell notation | \| |
| 3 | Symbol for the salt bridge in cell notation | \|\| |
| 4 | Side of the cell diagram where oxidation occurs | Left |
| 5 | Cell notation for the Zn/Cu cell | $Zn(s)\|Zn^{2+}(aq)\|\|Cu^{2+}(aq)\|Cu(s)$ |
| 6 | Half-cell that is the negative electrode | More negative $E^{\ominus}$ |
| 7 | Strongest oxidising agent | Most positive $E^{\ominus}$ (left side) |
| 8 | Strongest reducing agent | Most negative $E^{\ominus}$ (right side) |
| 9 | Sign of $E_{cell}$ for a feasible reaction | Positive |
| 10 | Why a feasible reaction may not happen | High $E_a$ or non-standard conditions |

# C20 Electrode Potentials and Cells – Knowledge Organiser

*AQA A-level Chemistry • Set 3: Commercial Cells* **Paper 1**

| # | Question | Answer |
|---|---|---|
| 1 | Why some cells can be recharged | Reactions are reversible |
| 2 | Lithium-ion cell: negative electrode half-equation | $Li \rightarrow Li^+ + e^-$ |
| 3 | Lithium-ion cell: positive electrode half-equation | $Li^+ + CoO_2 + e^- \rightarrow Li^+[CoO_2]^-$ |
| 4 | Fuel cell | Cell supplied continuously with fuel and oxygen |
| 5 | Alkaline hydrogen fuel cell: negative electrode | $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$ |
| 6 | Alkaline hydrogen fuel cell: positive electrode | $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ |
| 7 | Overall fuel cell equation | $2H_2 + O_2 \rightarrow 2H_2O$ |
| 8 | Advantage of hydrogen fuel cells | Only product is water |
| 9 | Disadvantage of hydrogen fuel cells | $H_2$ is flammable, hard to store, often made from fossil fuels |
| 10 | Why fuel cells do not need recharging | Reactants supplied continuously |